

Principles of Economics

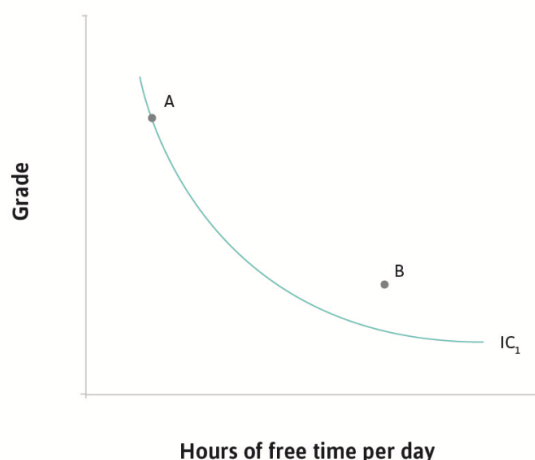
Problem Set 2

Universidad Carlos III de Madrid | Department of Economics

© 2026 Antonio Cabrales, Laurentiu Guinea Voinea, Ángel Hernando-Veciana, Bruno Pessoa and the Principios de Economía teaching team, Universidad Carlos III de Madrid. Licensed under CC BY-NC-SA 4.0, except where otherwise noted. Figures from CORE Econ, *The Economy 2.0*, under CC BY-NC-ND 4.0.

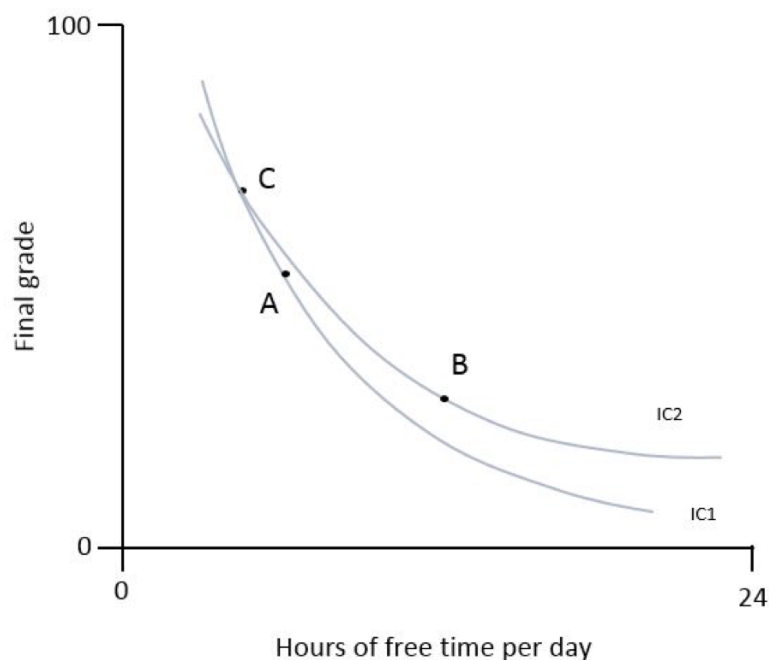
1. WARMING-UP QUESTIONS

1. Indicate ways in which you (as student) can improve the technology of your production function; that is, you can transform a certain number of study hours into more learning and thus higher grades. What role do you think your preferences play in allocating your scarce resource (time) to alternative uses?
2. In the diagram below, IC_1 is an indifference curve that depicts the preferences of a student. It shows all the possible combinations of grade obtained and free time enjoyed that give the same level of utility. As you can see, point A is on the indifference curve. The combination of grade and free time at point B is not on IC_1 .



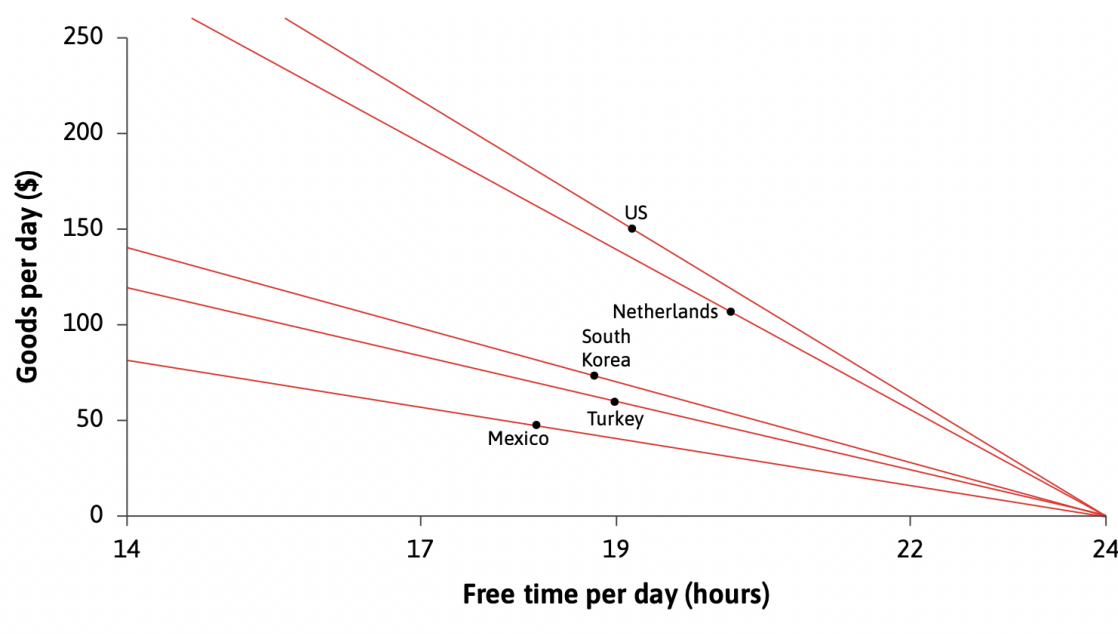
- a) Does combination at point B give higher or lower utility than combination at point A? How do you know?
- b) Draw a sketch of the diagram, and add another indifference curve, IC_2 , that goes through B and crosses IC_1 . Label the point at which IC_2 and IC_1 cross as point C.
- c) Combinations B and C are both on IC_2 . What does that imply about their levels of utility?
- d) Combinations C and A are both on IC_1 . What does that imply about their levels of utility?
- e) According to your answers to (3) and (4), how do the levels of utility at combinations A and B compare?

- f) Now compare your answers to (1) and (5). Explain how you know that two indifference curves that belong to the same map of indifference curves (they represent the same preferences) never cross.



3. Suppose that the points indicated in the following figure (Figure 3.25 of the text) reflect the choices of free time and consumption made by a typical worker in each of the five countries considered. You can find the data in OECD. Average annual hours actually worked per worker, and OECD. Household disposable income.

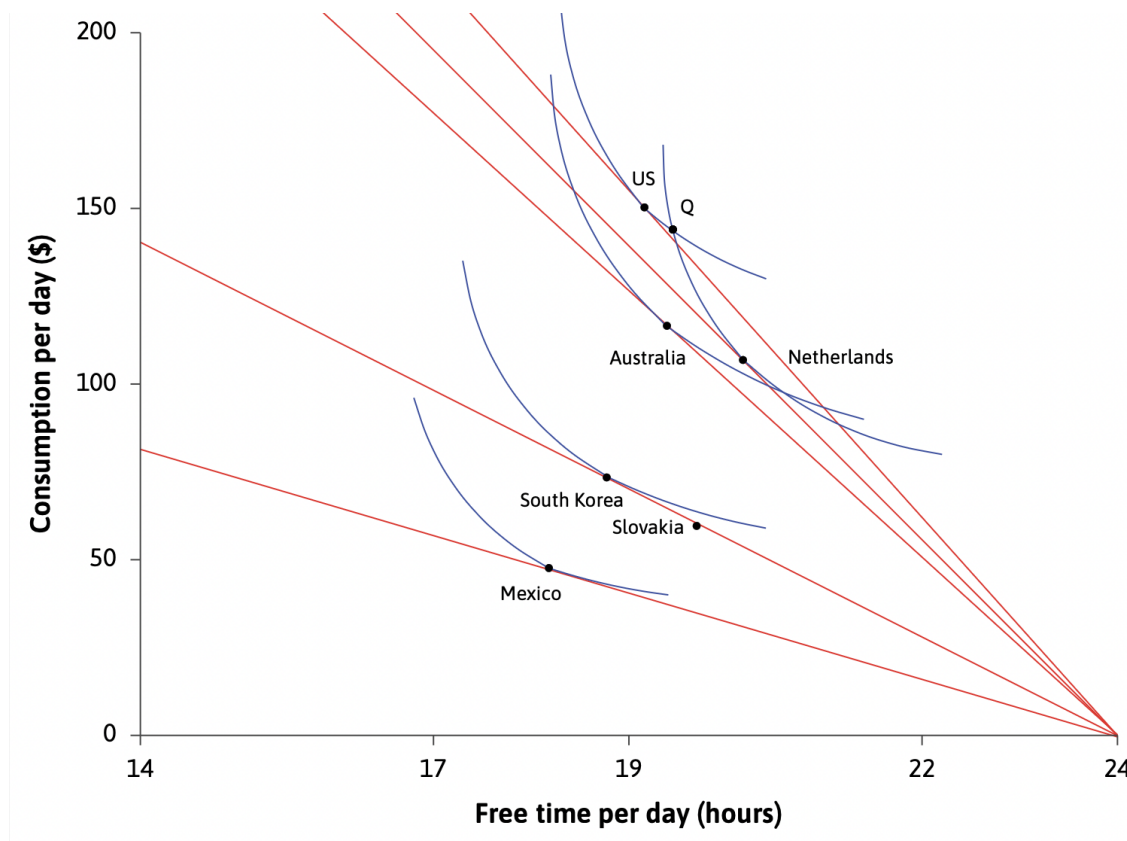
We take the wage (disposable income per hour worked), free time per day, and consumption per day to plot Figure 3.25 using our model. Notice that the wage is necessary to calculate the slope of the budget constrain for each country.



(To answer the following questions, you may need to draw indifference curves using a “representative worker” for each country, based on average free time and consumption per day)

in that country.)

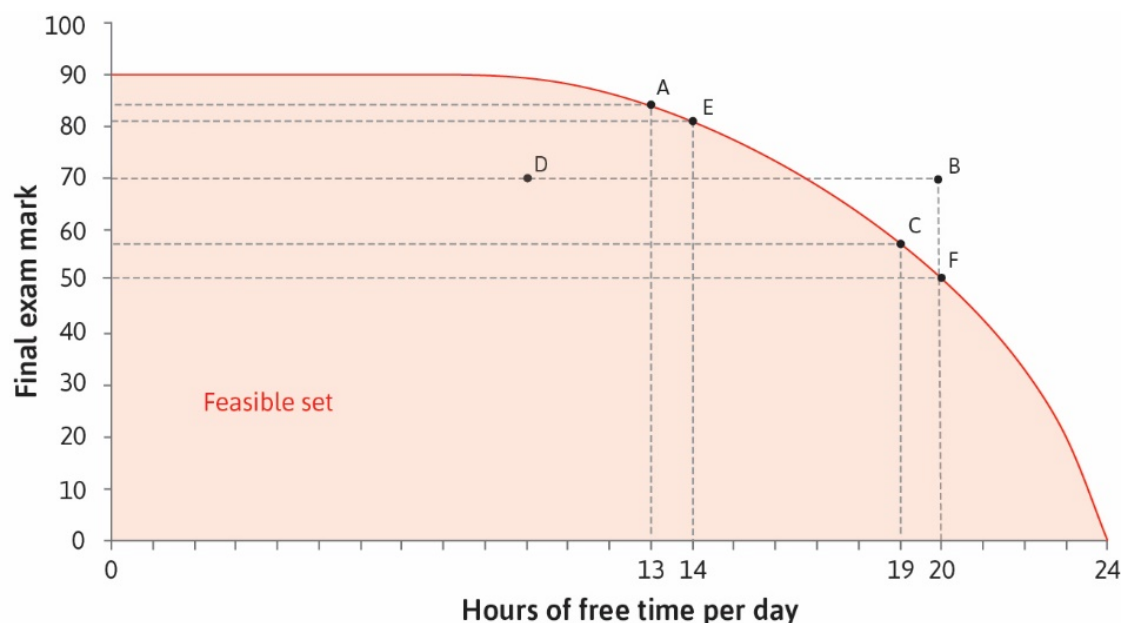
- Is it possible that people in Australia and the US have the same preferences? If so, how will a wage rise in Australia affect consumption and free time? What does this imply about the income and substitution effects?
- Suppose that people in Mexico and South Korea have the same preferences. In that case, what can you say about the income and substitution effects of a wage increase?



2. MULTIPLE CHOICE QUESTIONS

- Consider indifference curves for the consumption of two goods, milk and chocolate. The indifference curves are drawn with the number of chocolate bars on the horizontal axis and pints of milk on the vertical axis. Suppose consumer A has a flatter indifference curve than consumer B. Under these conditions, we can conclude that:
 - Consumer A likes chocolate more than consumer B.
 - The price of milk relative to the price of chocolates is higher for consumer A than for consumer B.
 - The indifference curves of the two consumers cannot cross.
 - If in the starting point the two individuals have the same quantity of chocolate bars, we can assert that consumer A is willing to swap one bar of chocolate for a smaller amount of milk than consumer B.
- The following diagram is the feasible set of a student. It shows combinations of her final grade and hours of free time per day. We can study her behavior by discussing what happens at

points A, B, C, D, E and F, which in principle are choices she can make given her preferences. Notice that given her preferences, she can only maximize utility at a single point on the feasible frontier. Moving between two points on the same feasible frontier (accepting the opportunity cost it entails) can only take place if there is a change in preferences. Keeping all these considerations in mind, only one of the following is correct:



	A	E	C	F
Free time	13	14	19	20
Grade	84	81	57	50
Opportunity cost	3		7	

- If she has to choose between B, C and F, she will choose to be at point B because she can get a higher grade for the same hours of free time compared to F, and compared to C she gets more of both goods.
- If she is at D, she can gain a lot by moving to B.
- If she compares A and C, no matter what her preferences are, she will choose C because she enjoys a lot more of free time and still gets a good mark.
- Going from A to E (one additional hour of free time) costs her 3 points in her mark (the opportunity cost). However, going from C to F costs her 7 points. Whether she is at A or C depends on her preferences.

3. NUMERICAL PROBLEMS

- You work 40 hours per week for a wage of 20 euros an hour. Your free hours are defined as the number of hours per week not spent working: $(24 \text{ hours} \times 7 \text{ days}) - 40 \text{ hours} = 128 \text{ hours}$ per week. Suppose now that your wage rate increases from 20 euros to 25 euros. However, you are perfectly happy keeping your current level of consumption, that is, you don't want to change it. How many hours will you be working after the wage rise? In a graph, explain your behavior using our model of time allocation between work and free time. Consider the income and substitution effects to make sense of your answer.

4. CONCEPTUAL QUESTIONS

Write down a short and concise answer. When you are asked to solve the question in class, explain the concept clearly and give examples or pieces of evidence.

7. In “An Essay on the Nature and Significance of Economic Science”, published in 1932, Lionel Robbins wrote: ‘Economics is the science that studies human behaviour as a relationship between given ends and scarce means which have alternative uses.’ Although this is the best-known definition of Economics, do you think that economists are concerned only with the allocation of scarce resources, given their alternative uses? Can we say that labour is a scarce resource if 15% of the labour force is unemployment?
8. Do our models of scarcity and choice provide a plausible explanation for the observed trends in working hours during the twentieth century? What other factors, not included in our model, might be important in explaining what has happened? Remember Keynes’ prediction that working hours would fall to 15 hours per week in the century after 1930. Why do you think working hours have not changed as he expected? Have people’s preferences changed? The model focuses on the number of hours workers would choose, so do you think that many employees are now working longer than they would like? Remember that our model tells us that people’s best (utility-maximizing) choice is the level of working hours at which the slope of the feasible frontier is equal to the slope of the indifference curve. Keep also in mind that an increase in wages leads to both an income effect and a substitution effect. Which one dominates depends on the budget constraint and preferences.